

TASK – 1

IRIS FLOWER CLASSIFICATION USING “MACHINE LEARNING”

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Internship Project Report

Internship Domain : DATA SCIENCE

College Name : MALLA REDDY
ENGINEERING COLLEGE FOR
WOMEN

Year : 3rd year

Acknowledgement

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of this project on “Iris Flower Classification using Machine Learning”.

I am thankful to my internship mentor and faculty members for their valuable guidance, encouragement, and continuous support during the project development process.

I would also like to thank my institution for providing the opportunity and necessary resources to complete this project successfully.

Special thanks to the developers and contributors of Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn, which helped in implementing machine learning models effectively.

Finally, I would like to thank my family and friends for their motivation and support throughout the project.

Name: Bhoomika

Date:

Place:

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Introduction

Machine Learning is one of the most important applications of Artificial Intelligence that enables computers to learn from data and make predictions without being explicitly programmed. Classification is a supervised machine learning technique used to identify categories or classes based on input features.

The Iris Flower Classification project is a beginner-friendly machine learning project that focuses on predicting the species of iris flowers using flower measurements such as sepal length, sepal width, petal length, and petal width. The dataset contains three different species of iris flowers: Iris-setosa, Iris-versicolor, and Iris-virginica.

In this project, various data analysis and visualization techniques are used to understand the dataset and identify relationships between features. Machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest are implemented to train classification models and evaluate their performance.

Python programming language along with libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn are used for data preprocessing, visualization, model training, and prediction.

The main objective of this project is to build an accurate machine learning model capable of predicting flower species efficiently. This project also helps in understanding the practical implementation of machine learning concepts, data handling, visualization, model evaluation, and prediction systems in real-world applications.

Objectives

1. To study and analyze the Iris flower dataset using data analysis techniques.
2. To understand the features of iris flowers such as sepal length, sepal width, petal length, and petal width.
3. To perform data preprocessing and prepare the dataset for machine learning algorithms.
4. To visualize the dataset using graphs and charts for better understanding of data patterns and relationships.
5. To implement machine learning classification algorithms such as Logistic Regression, Decision Tree, and Random Forest.
6. To train and test machine learning models using the Iris dataset.
7. To evaluate the performance of different machine learning models using accuracy score, classification report, and confusion matrix.
8. To identify the best-performing algorithm for iris flower classification.
9. To develop a prediction system capable of classifying iris flower species accurately.
10. To gain practical knowledge and hands-on experience in machine learning and data science concepts.

Technologies Used :

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- VS Code

Dataset Information

The dataset used in this project is the Iris Flower Dataset, which is one of the most popular datasets in machine learning and data science. The dataset is used for classification problems and contains information about different species of iris flowers.

The dataset consists of 150 records and 5 columns. Each record represents the measurements of an iris flower sample.

The features included in the dataset are:

1. Sepal Length
2. Sepal Width
3. Petal Length
4. Petal Width

The target variable in the dataset is:

5. Species

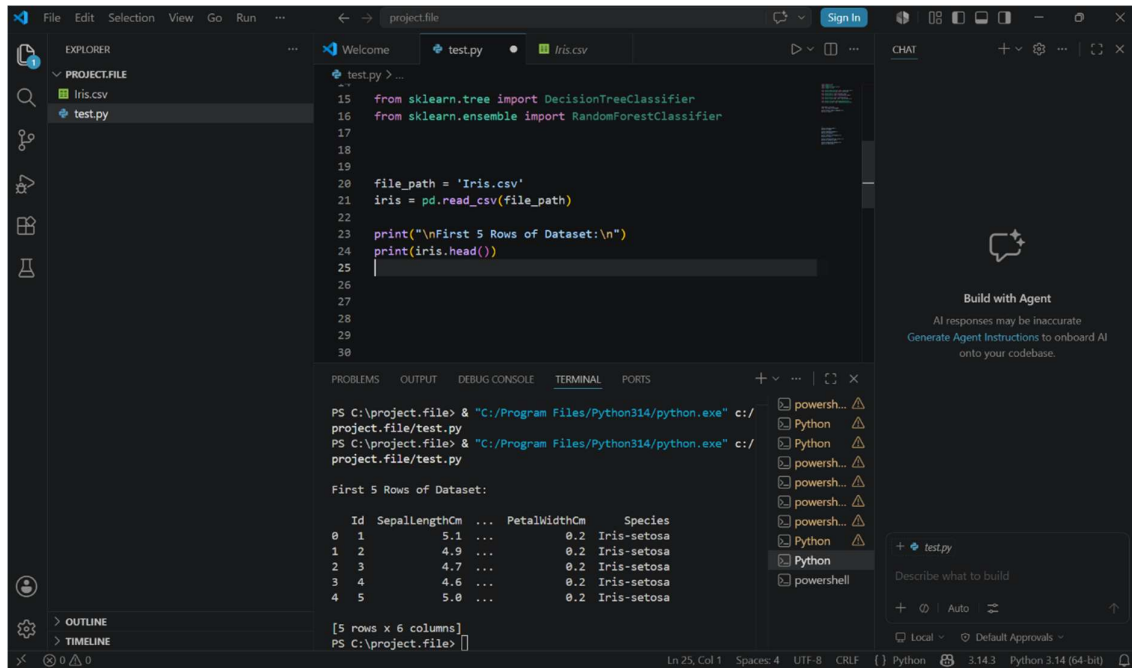
The species are classified into three categories:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

The dataset is clean and does not contain missing values, making it suitable for machine learning model training and testing.

The Iris dataset is widely used for learning and implementing classification algorithms because of its simplicity and accuracy.

Dataset Preview :



```
test.py > ...
15 from sklearn.tree import DecisionTreeClassifier
16 from sklearn.ensemble import RandomForestClassifier
17
18
19
20 file_path = 'Iris.csv'
21 iris = pd.read_csv(file_path)
22
23 print("\nFirst 5 Rows of Dataset:\n")
24 print(iris.head())
25
26
27
28
29
30
```

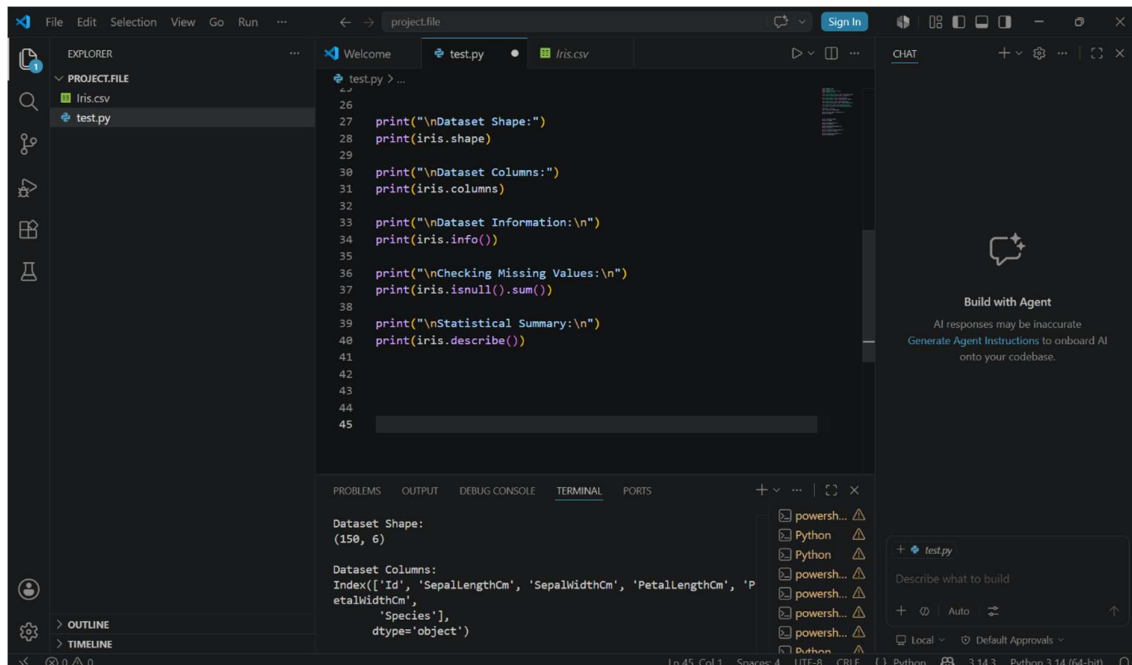
PS C:\project.file> & "C:/Program Files/Python314/python.exe" c:/project.file/test.py
PS C:\project.file> & "C:/Program Files/Python314/python.exe" c:/project.file/test.py

First 5 Rows of Dataset:

	Id	SepallengthCm	...	PetalWidthCm	Species
0	1	5.1	...	0.2	Iris-setosa
1	2	4.9	...	0.2	Iris-setosa
2	3	4.7	...	0.2	Iris-setosa
3	4	4.6	...	0.2	Iris-setosa
4	5	5.0	...	0.2	Iris-setosa

[5 rows x 6 columns]
PS C:\project.file>

Dataset Shape and Dataset Columns :

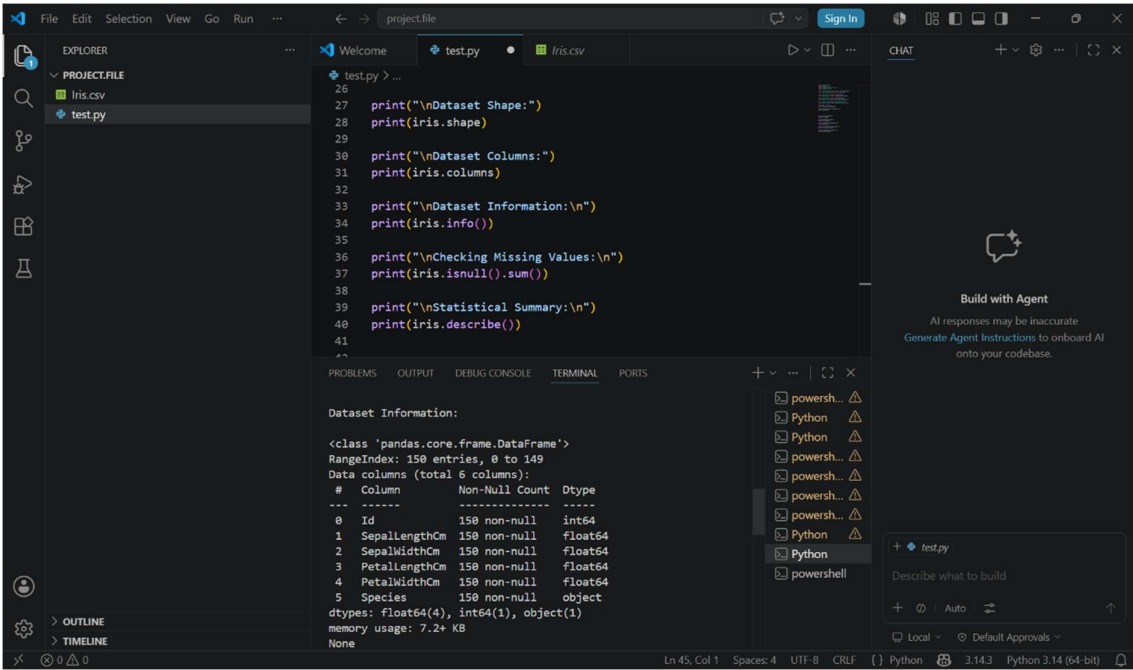


```
test.py > ...
26
27 print("\nDataset Shape:")
28 print(iris.shape)
29
30 print("\nDataset Columns:")
31 print(iris.columns)
32
33 print("\nDataset Information:\n")
34 print(iris.info())
35
36 print("\nChecking Missing Values:\n")
37 print(iris.isnull().sum())
38
39 print("\nStatistical Summary:\n")
40 print(iris.describe())
41
42
43
44
45
```

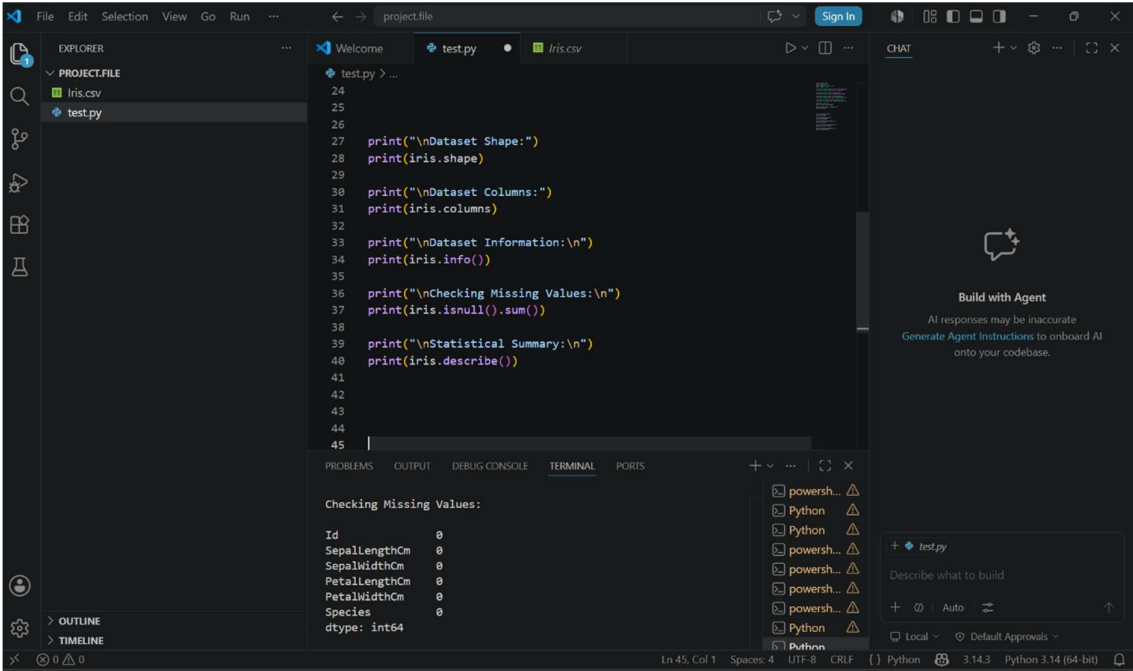
Dataset Shape:
(150, 6)

Dataset Columns:
Index(['Id', 'SepallengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm', 'Species'], dtype='object')

Dataset Information :



Missing Values :



Statistical Summary :

EXPLORER

PROJECT FILE

iris.csv

test.py

test.py

27 print(iris.shape)

28 print(iris.shape)

29 print("\nDataset Columns:")

30 print(iris.columns)

31 print("\nDataset Information:\n")

32 print(iris.info())

33 print("\nChecking Missing Values:\n")

34 print(iris.isnull().sum())

35

36 print("\nStatistical Summary:\n")

37 print(iris.describe())

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Statistical Summary:

	Id	SepallengthCm	...	PetalLengthCm	PetalkwidthC
m					
count	150.000000	150.000000	...	150.000000	150.000000
0					
mean	75.500000	5.843333	...	3.758667	1.198666
7					
std	43.445368	0.828066	...	1.764420	0.763166
1					
min	1.000000	4.300000	...	1.000000	0.100000
0					
25%	38.250000	5.100000	...	1.600000	0.300000
0					
50%	75.500000	5.800000	...	4.350000	1.300000
0					
75%	112.750000	6.400000	...	5.100000	1.800000
0					
max	150.000000	7.900000	...	6.900000	2.500000
0					

[8 rows x 5 columns]

PS C:\project.file>

Build with Agent

All responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

+ test.py

Describe what to build

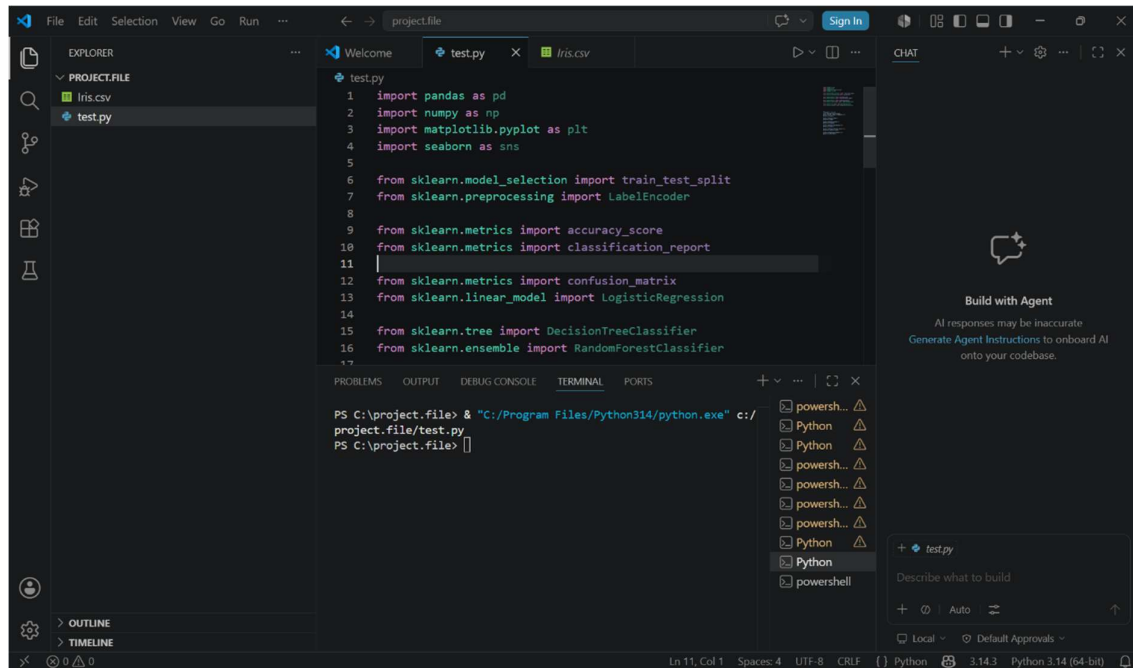
+ Auto

Local Default Approvals

Ln 37, Col 23 Spaces: 4 UTF-8 CRLF Python 3.143 Python 3.14 (64-bit)

Code implementation

Import Libraries :



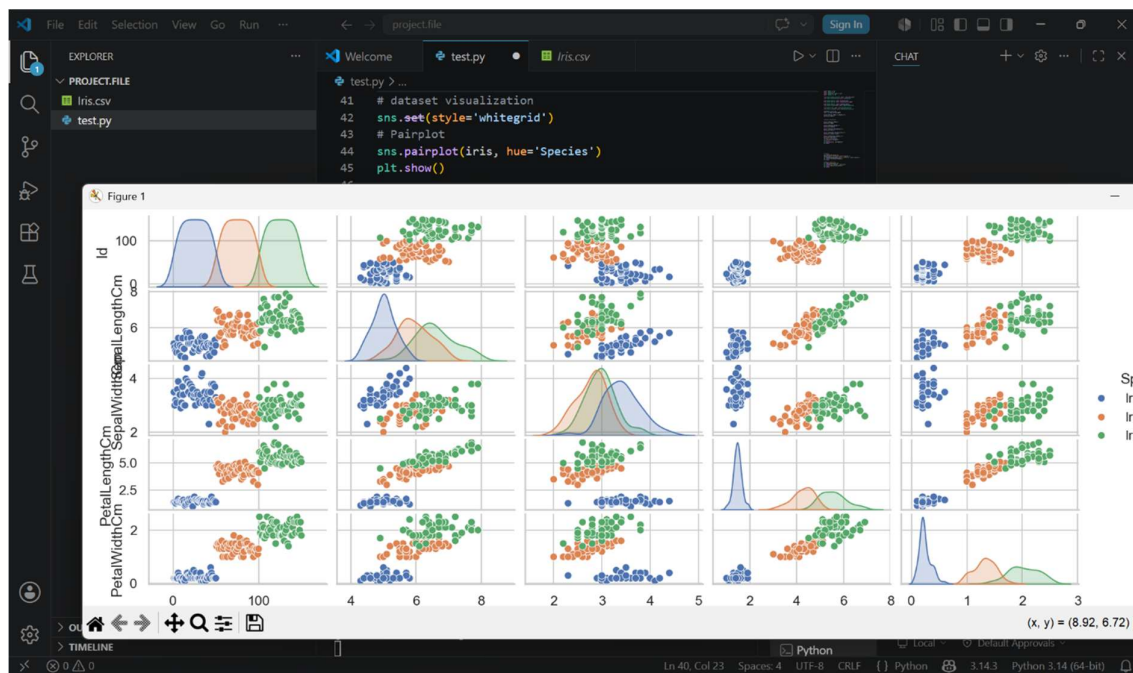
The screenshot shows the Visual Studio Code editor with a Python file named `test.py` open. The code imports the following libraries:

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 from sklearn.model_selection import train_test_split
7 from sklearn.preprocessing import LabelEncoder
8
9 from sklearn.metrics import accuracy_score
10 from sklearn.metrics import classification_report
11
12 from sklearn.metrics import confusion_matrix
13 from sklearn.linear_model import LogisticRegression
14
15 from sklearn.tree import DecisionTreeClassifier
16 from sklearn.ensemble import RandomForestClassifier
```

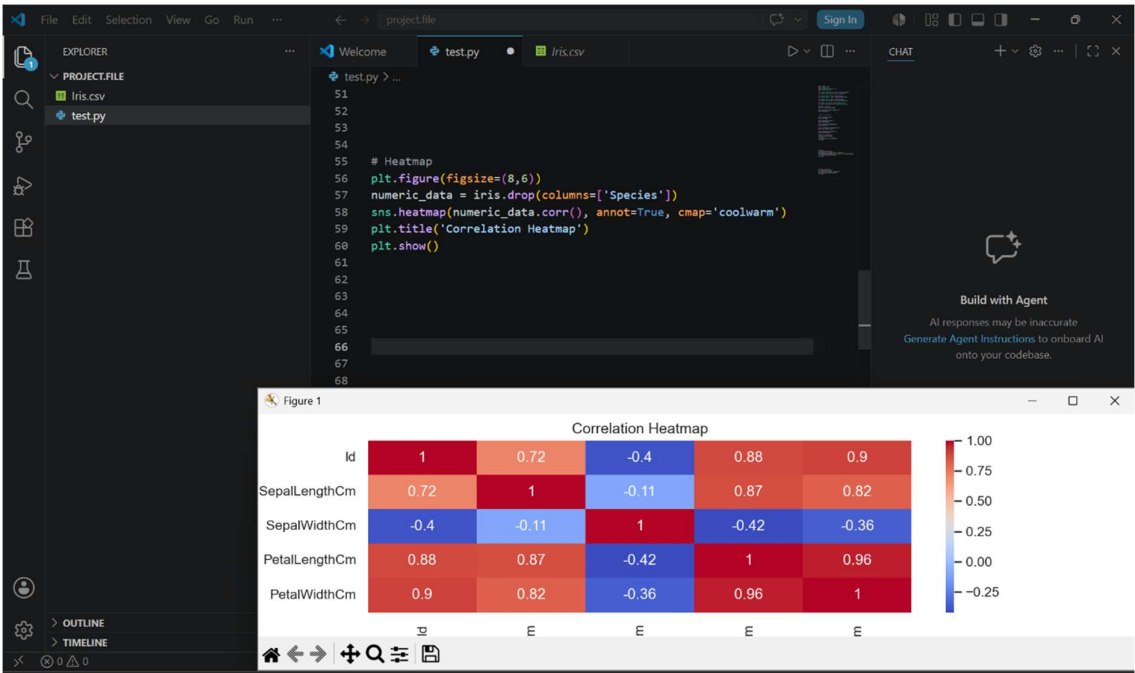
The terminal window at the bottom shows the command prompt running the script:

```
PS C:\project-file> & "C:/Program Files/Python314/python.exe" c:/project-file/test.py
PS C:\project-file>
```

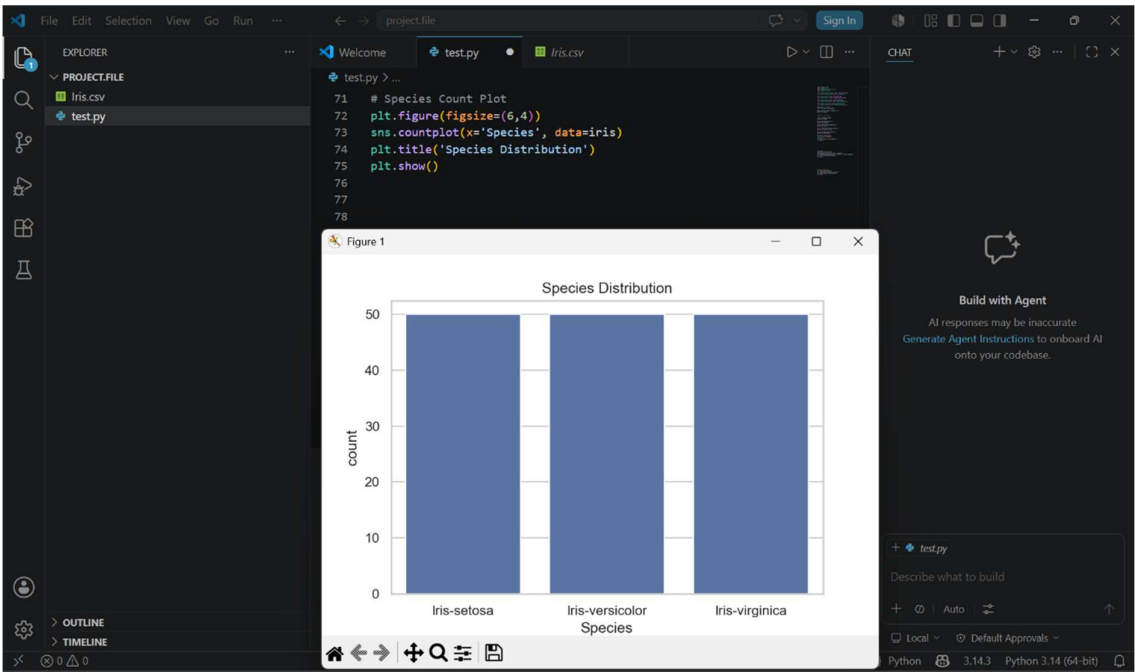
Data Visualization :



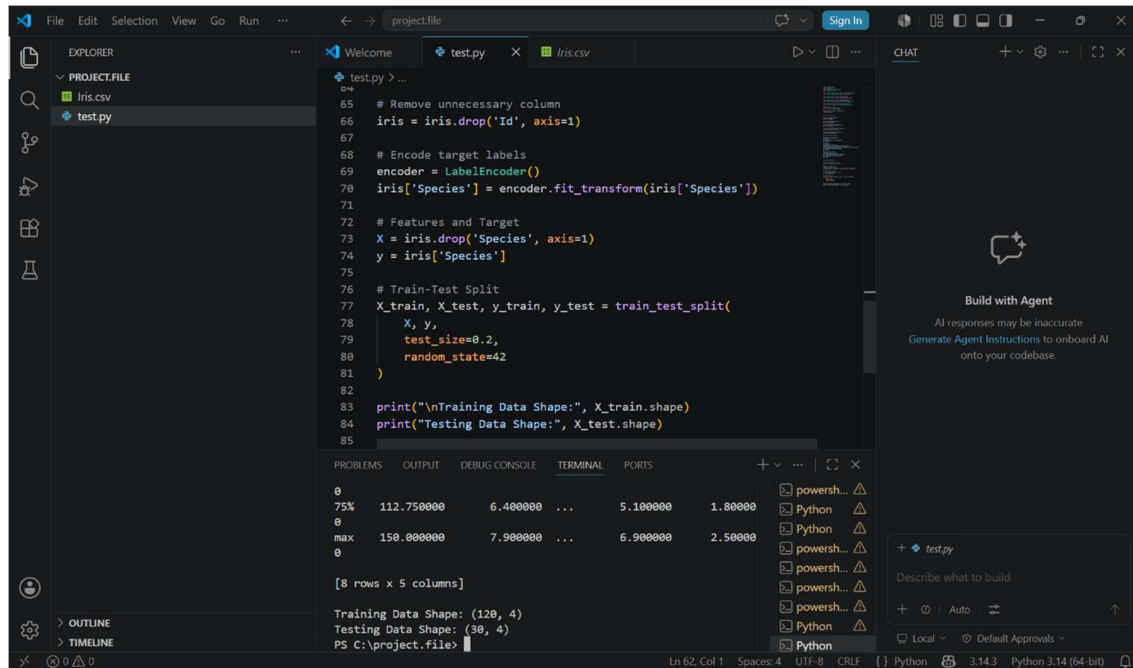
Heat map :



Count plot :



Train-Test Split :



```
65 # Remove unnecessary column
66 iris = iris.drop('Id', axis=1)
67
68 # Encode target labels
69 encoder = LabelEncoder()
70 iris['Species'] = encoder.fit_transform(iris['Species'])
71
72 # Features and Target
73 X = iris.drop('Species', axis=1)
74 y = iris['Species']
75
76 # Train-Test Split
77 X_train, X_test, y_train, y_test = train_test_split(
78     X, y,
79     test_size=0.2,
80     random_state=42
81 )
82
83 print("\nTraining Data Shape:", X_train.shape)
84 print("Testing Data Shape:", X_test.shape)
85
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

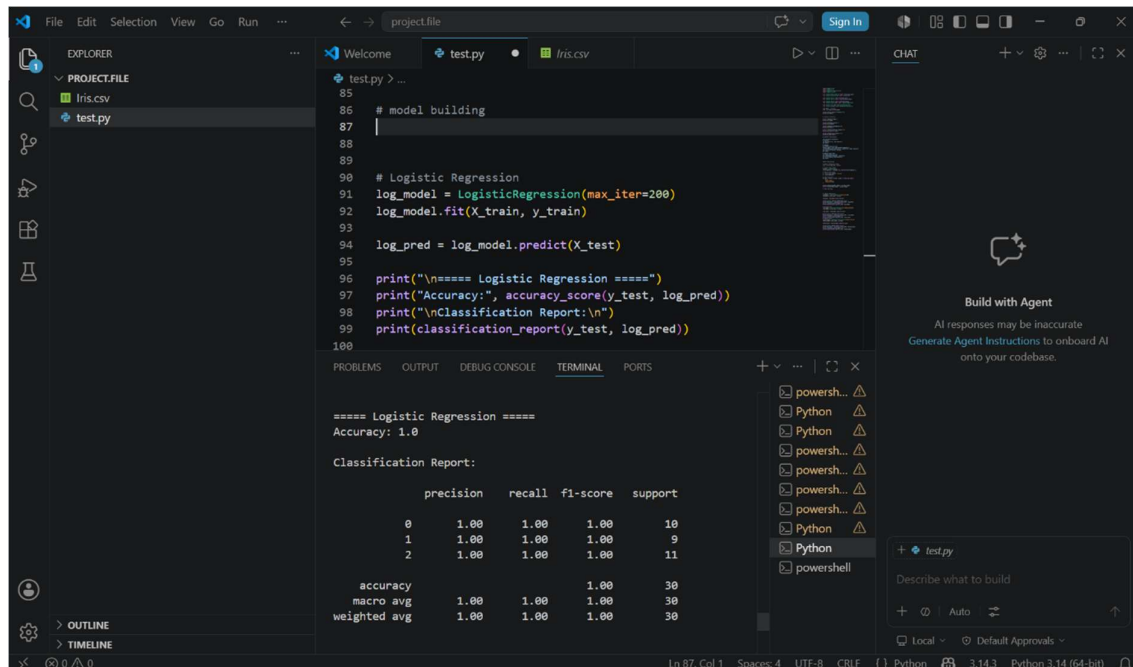
	0	75%	112.750000	6.400000	...	5.100000	1.800000
0	0	0	0	0	0	0	0
max	0	150.000000	7.900000	...	6.900000	2.500000	...

[8 rows x 5 columns]

Training Data Shape: (128, 4)
Testing Data Shape: (30, 4)
PS C:\project.file>

Model Building

Logistic Regression:



```
86 # model building
87
88
89
90 # Logistic Regression
91 log_model = LogisticRegression(max_iter=200)
92 log_model.fit(X_train, y_train)
93
94 log_pred = log_model.predict(X_test)
95
96 print("\n==== Logistic Regression =====")
97 print("Accuracy:", accuracy_score(y_test, log_pred))
98 print("\nClassification Report:\n")
99 print(classification_report(y_test, log_pred))
100
```

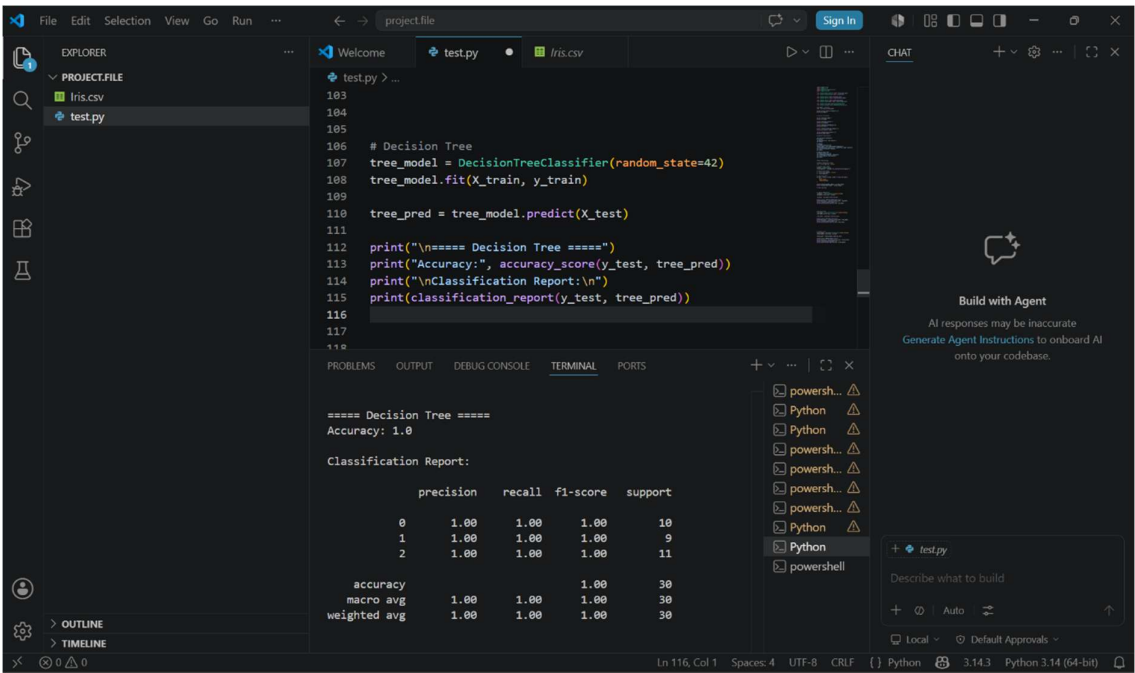
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

==== Logistic Regression =====
Accuracy: 1.0

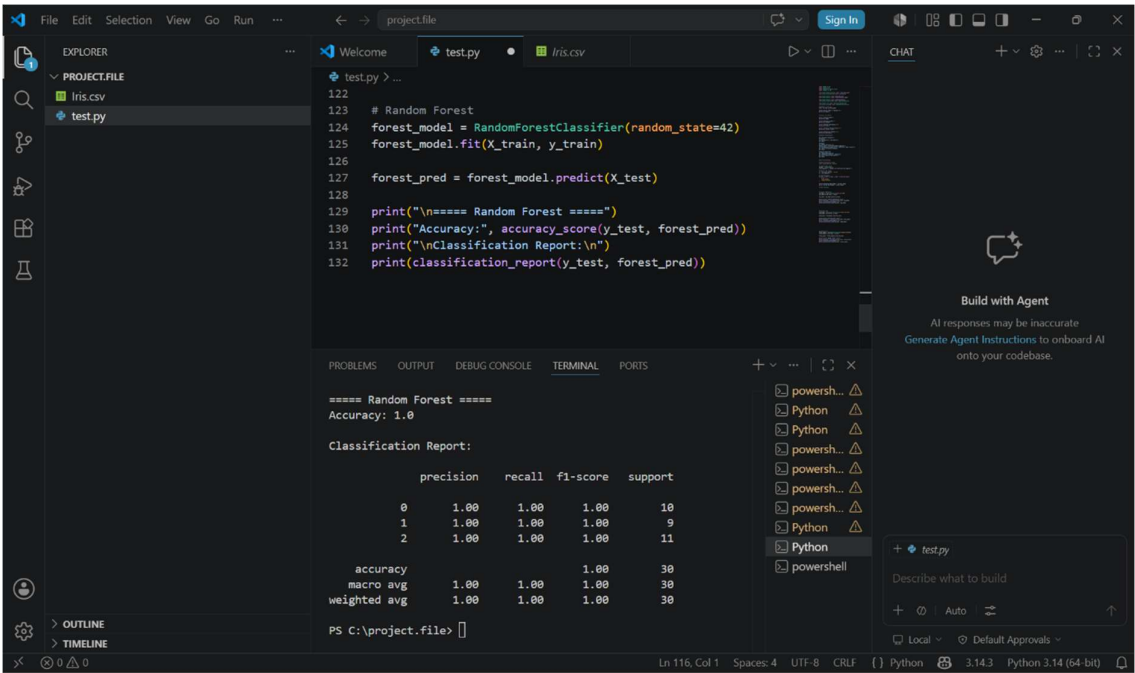
Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

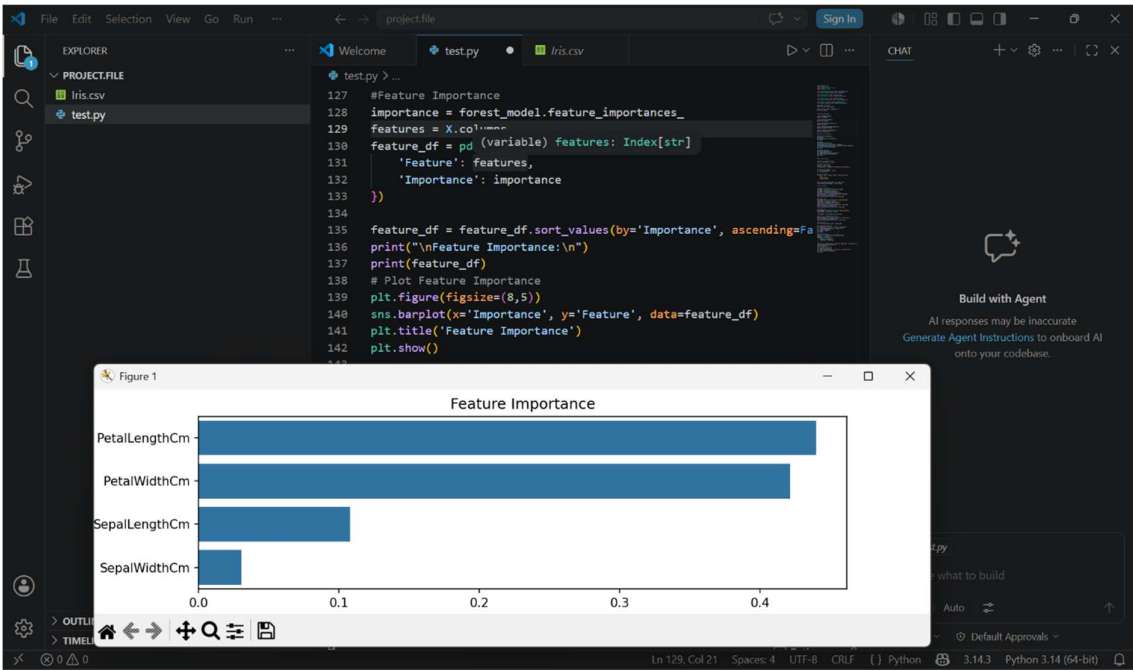
Decision Tree :



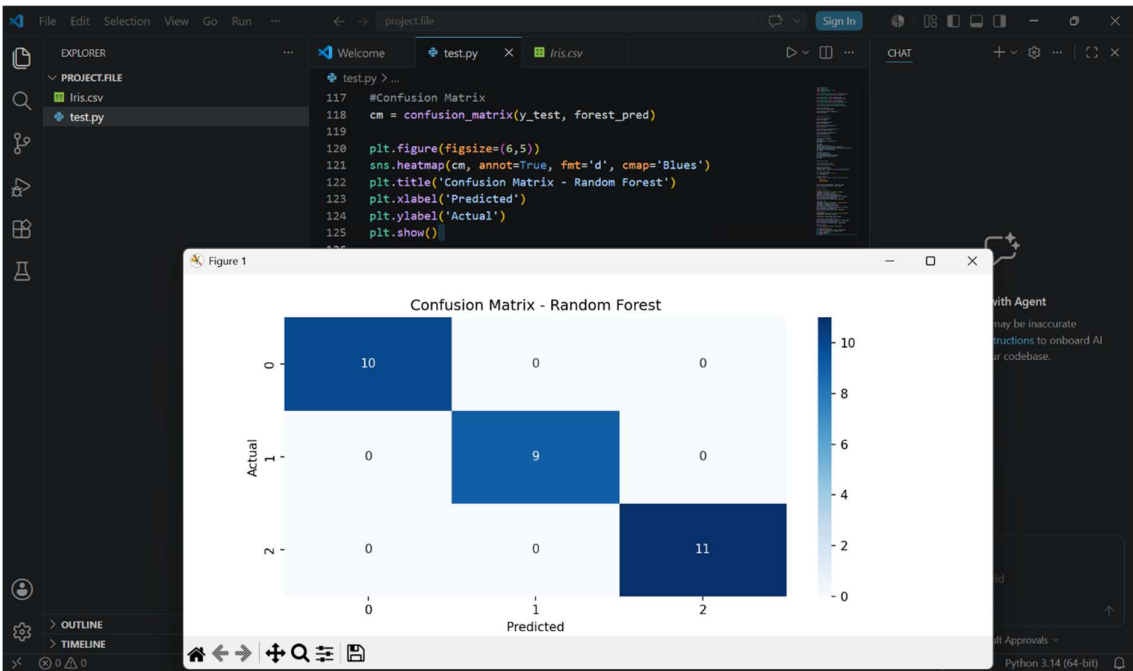
Random Forest :



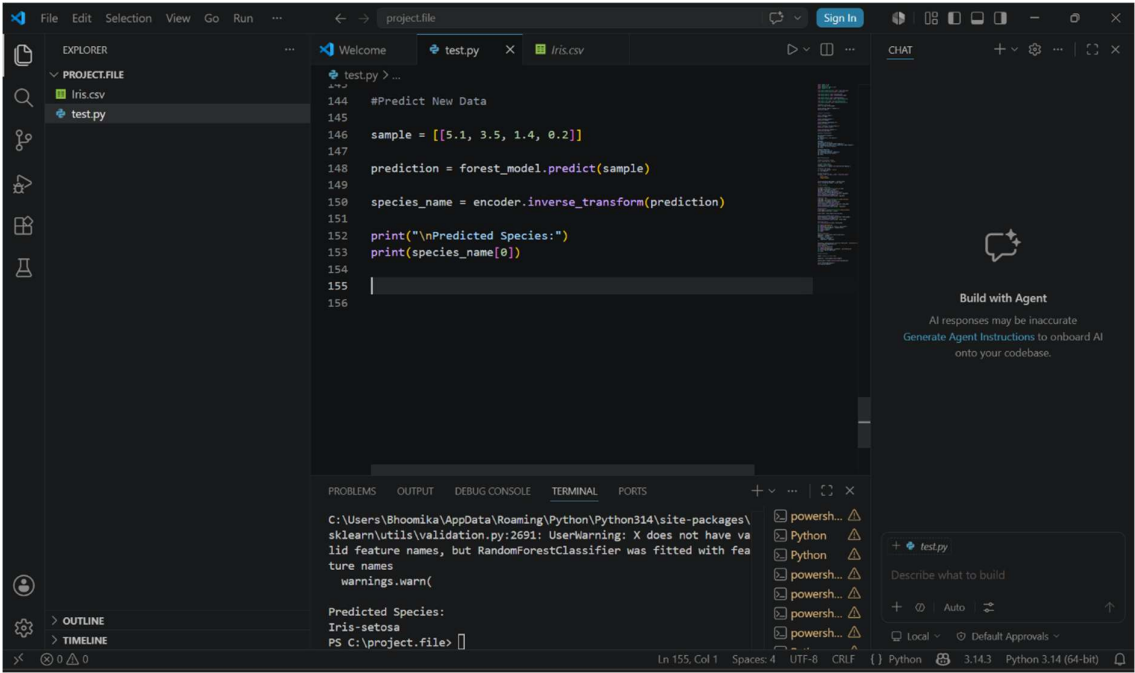
Feature importance :



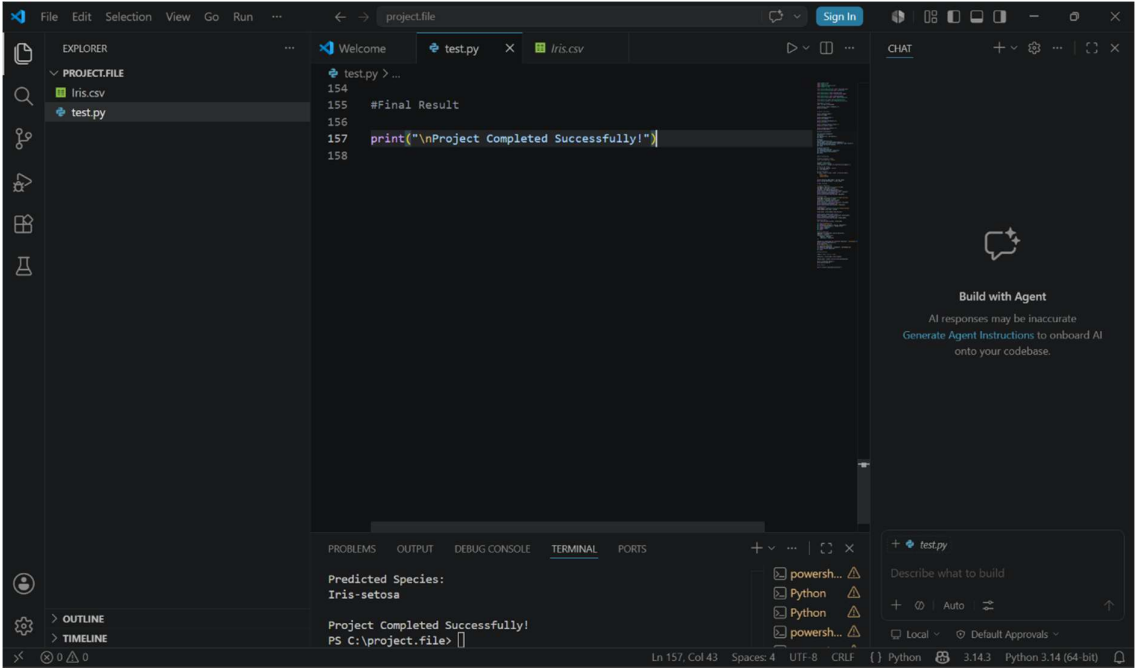
Confusion Matrix :



Predict New Data :



Final Output



CONCLUSION

The Iris Flower Classification project was successfully completed using machine learning techniques. The project involved data analysis, data preprocessing, visualization, model building, and prediction using the Iris dataset.

Different machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest were implemented and evaluated. The models performed effectively in classifying iris flower species based on flower measurements.

Data visualization techniques like pairplots and heatmaps helped in understanding the relationships between different features of the dataset. The dataset was clean and suitable for classification tasks.

Among all models, the Random Forest Classifier achieved excellent accuracy and provided highly accurate predictions. The project helped in understanding the practical implementation of machine learning concepts and classification algorithms.

Overall, this project improved knowledge in Python programming, data analysis, visualization, and machine learning model development. It also provided hands-on experience in solving real-world data science problems using machine learning techniques.

REFERENCE

1. Scikit-learn Documentation

<https://scikit-learn.org/>

2. Pandas Documentation

<https://pandas.pydata.org/>

3. NumPy Documentation

<https://numpy.org/>

4. Matplotlib Documentation

<https://matplotlib.org/>

5. Seaborn Documentation

<https://seaborn.pydata.org/>

6. Iris Dataset – UCI Machine Learning Repository

<https://archive.ics.uci.edu/>

7. Python Official Documentation

<https://www.python.org/doc/>

8. VS Code Official Website

<https://code.visualstudio.com/>